

Chapter 3 | Statistical Inference

Learning objectives

- Utilize the least squares (LS) estimator $\hat{\beta}$ and its standard errors to produce confidence intervals and hypothesis tests for β_j values.
- Convert hypothesis tests for $\beta_j = 0$ into a model comparison F -test.
- Utilize F -tests to perform a hypothesis test of multiple β_j values all being equal to zero. This is the simple vs. complex model comparison.
- Create confidence intervals for β_j values leveraging the normality assumption of residuals.

Outline

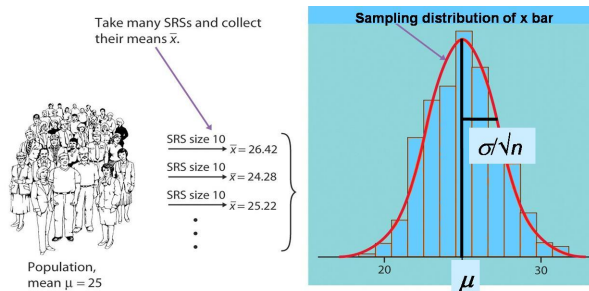
- 1 Introduction
- 2 Confidence Intervals and Hypothesis Tests
- 3 Comparison of Nested Models
- 4 Example

Sampling distributions

- Suppose **all possible samples** of a **fixed** sample size n are taken from a population and a sample statistic of interest, such as $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, is calculated from each sample.
- The **sampling distribution** of the statistic is the distribution of **all possible values** of the sample statistic.
- Statistical inference is based on the sampling distribution.
- Notes:
 - ① In practice, a sample of size n is taken only **once**, and only **one** value of the sample statistic is calculated.
 - ② A sampling distribution is therefore not an empirically observed distribution. Rather, it is a **theoretical** concept – we do not actually build it.

Sampling distribution of $\bar{X}$ | Sketch

- Suppose **all possible samples** of a **fixed** sample size n are taken from a population with mean μ and variance σ^2 and $\bar{X}$ is calculated from each sample.
 - Some $\bar{X}$ will be greater than the population mean μ and some $\bar{X}$ will be smaller, making up the sampling distribution.
 - These $\bar{X}$'s have a normal distribution centered at μ with standard deviation $\frac{\sigma}{\sqrt{n}}$.



Sampling distribution of $\bar{X}$ | Theory

- Suppose $X_1, \dots, X_n$ is a random sample of size n from a population with mean μ and variance σ^2 .
- The sampling distribution of $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ has
 - ① a **normal distribution** if n is sufficiently large
 - ② theoretical mean (expected value) $\mu_{\bar{X}}$ equal to the population mean μ : $\mu_{\bar{X}} = E(\bar{X}) = \mu$
 - ③ theoretical variance $\sigma_{\bar{X}}^2$ equal to the population variance σ^2 divided by the sample size n :
$$\sigma_{\bar{X}}^2 = \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$
- **Central Limit Theorem (CLT)**: Let $X_1, \dots, X_n$ be a random sample of size n from a population with mean μ and variance σ^2 . Then,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

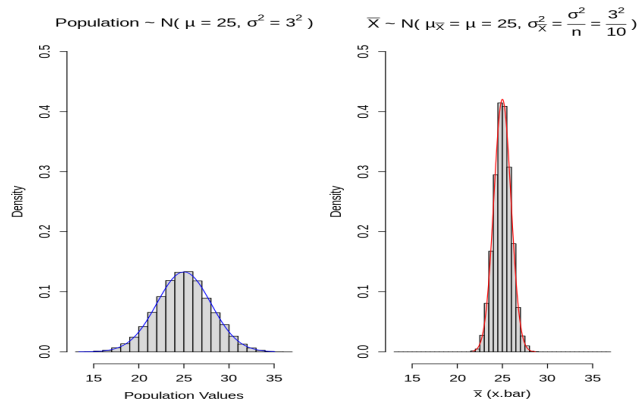
Sampling distribution of $\bar{X}$ | Simulation

- R scripts to generate a sampling distribution of $\bar{X}$:

```
1 mu = 25;  sigma = 3           # population mean and standard deviation
2 n = 10;           # sample size
3 n.rpt = 5000       # number of repetitions
4
5 x.bar = numeric( length=n.rpt ) # an empty vector for n.rpt sample means
6 set.seed( 12345 )  # a seed value for random number generation
7                   # to reproduce the same result
8 for ( k in 1:n.rpt ) {        # repeat the following procedures n.rpt times
9   x.smp = rnorm( n, mean=mu, sd=sigma ) # generate a sample from N(mu,sigma^2)
10  x.bar[k] = mean( x.smp )      # compute & keep the sample mean for each sample
11 }
```

Population distribution vs. Sampling distribution of $\bar{X}$

- Population distribution (left) vs. Simulated sampling distribution of $\bar{X}$ (right)



- We use this $\bar{X}$ -distribution to perform statistical inference on the population mean μ .

Confidence interval for μ

- A 95% confidence interval for the population mean μ is

$$\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

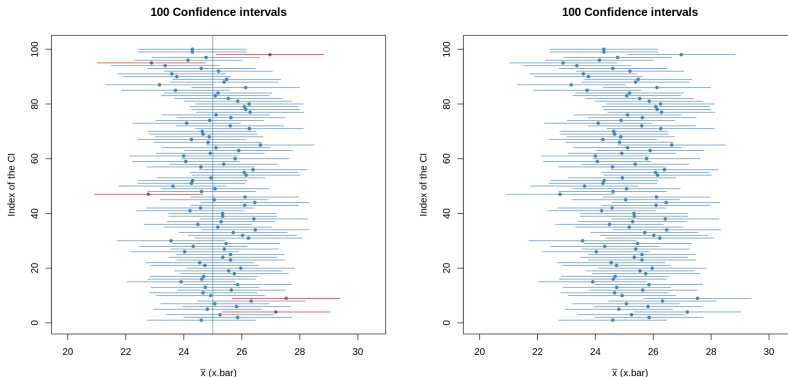
- 1 Why? Because $\bar{X} \sim N(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n})$, we have

$$P\left(-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right) = 0.95 \Rightarrow P\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}}\right) = 0.95$$

- 2 We assumed σ is known.
- 3 Otherwise, we will replace σ with the sample standard deviation s and 1.96 with an equivalent quantile from a t -distribution with $n - 1$ degrees of freedom (if n is small or moderate).

Confidence interval for μ

- Confidence intervals: $\mu = 25$ is **known** (left) vs. μ is **unknown** (right)



- 1 If μ is **known**, 95% of these CIs contain μ , and in addition, we know CIs that contain μ .
- 2 In practice, μ is **unknown**. Therefore, we cannot tell whether a particular CI contains unknown μ or not. However, the percentage of CIs that contain unknown μ remains the same: 95%.

Sampling distribution of $\hat{\beta}$ in SLR | Theory

- Suppose $(X_1, Y_1), \dots, (X_n, Y_n)$ are a random sample of size n for SLR: $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$, where ϵ_i 's are IID $N(0, \sigma^2)$.
- The sampling distribution of LS estimator $\hat{\beta}_1$ has
 - ① a **normal distribution** if n is sufficiently large
 - ② theoretical mean (expected value) $\mu_{\hat{\beta}_1}$ equal to the slope parameter β_1 : $\mu_{\hat{\beta}_1} = E(\hat{\beta}_1) = \beta_1$
 - ③ theoretical variance $\sigma_{\hat{\beta}_1}^2$ equal to the model error variance σ^2 times $(\mathbf{X}^T \mathbf{X})_{[2,2]}^{-1}$:
 $\sigma_{\hat{\beta}_1}^2 = \text{Var}(\hat{\beta}_1) = \sigma^2 (\mathbf{X}^T \mathbf{X})_{[2,2]}^{-1}$
 - ④ Note: If there are no predictors, then $\mathbf{X} = [1, \dots, 1]^T$ and therefore $\mathbf{X}^T \mathbf{X} = n$.
- In summary,

$$\hat{\beta}_1 \sim N\left(\beta_1, \sigma^2 (\mathbf{X}^T \mathbf{X})_{[2,2]}^{-1}\right)$$

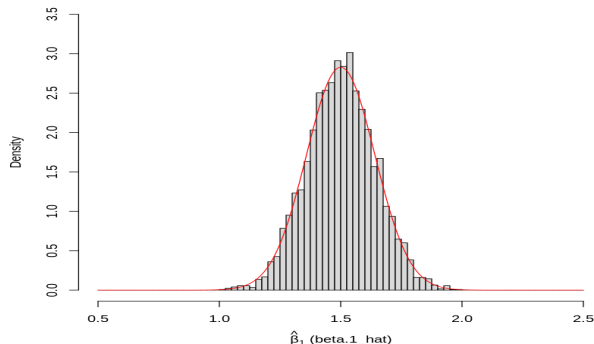
Sampling distribution of $\hat{\beta}_1$ in SLR | Simulation

- R scripts to generate a sampling distribution of $\hat{\beta}_1$ in SLR:

```
1 b.0 = 10; b.1 = 1.5; sigma = 2      # SLR coef's and error standard deviation
2 n = 100                             # sample size
3 n.rpt = 5000                        # number of repetitions
4
5 set.seed( 123 )                     # a seed value for reproducible results
6 x = runif( n, min=0, max=5 )        # generate x_1,...,x_n from Uniform(0,5)
7
8 b.1_est = numeric( length=n.rpt )   # an empty vector for n.rpt b.1 slope estimates
9 set.seed( 2345 )                   # a seed value for reproducible results
10 for ( k in 1:n.rpt ) {              # repeat the following procedures n.rpt times
11   e = rnorm( n, mean=0, sd=sigma )  # generate errors e_1,...,e_n from N(0,sigma^2)
12   y = b.0 + b.1*x + e               # generate responses y_1,...,y_n
13   fit = lm( y ~ x )                 # fit SLR model
14   b.1_est[k] = coef( fit )[2]       # keep b.1 LS estimate for each sample
15 }
```

Sampling distribution of $\hat{\beta}_1$ in SLR | Simulation

- Simulated sampling distribution of $\hat{\beta}_1$: $\hat{\beta}_1 \sim N\left(\beta_1, \sigma^2(\mathbf{X}^T\mathbf{X})_{[2,2]}^{-1}\right)$



- We use this $\hat{\beta}_1$ -distribution to perform statistical inference on the slope parameter β_1 .

Confidence interval for β_1

- A 95% confidence interval for the population mean $\hat{\beta}_1$ is

$$\left(\hat{\beta}_1 - 1.96\sigma\sqrt{(\mathbf{X}^T\mathbf{X})_{[2,2]}^{-1}}, \hat{\beta}_1 + 1.96\sigma\sqrt{(\mathbf{X}^T\mathbf{X})_{[2,2]}^{-1}} \right)$$

- Why? Because $\hat{\beta}_1 \sim N\left(\beta_1, \sigma^2(\mathbf{X}^T\mathbf{X})_{[2,2]}^{-1}\right)$, we have

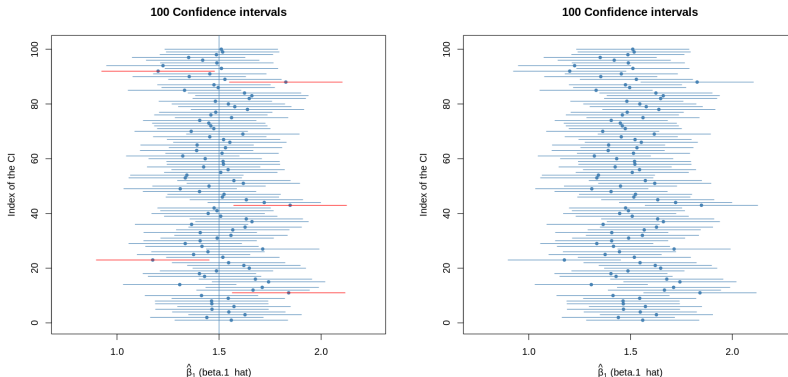
$$P\left(-1.96 < \frac{\hat{\beta}_1 - \beta_1}{\sigma\sqrt{(\mathbf{X}^T\mathbf{X})_{[2,2]}^{-1}}} < 1.96\right) = 0.95$$

$$\Rightarrow P\left(\hat{\beta}_1 - 1.96\sigma\sqrt{(\mathbf{X}^T\mathbf{X})_{[2,2]}^{-1}} < \beta_1 < \hat{\beta}_1 + 1.96\sigma\sqrt{(\mathbf{X}^T\mathbf{X})_{[2,2]}^{-1}}\right) = 0.95$$

- We assumed σ is known.
- Otherwise, we will replace σ with the square root of the mean squared errors (MSE)
 $\hat{\sigma} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$ and 1.96 with an equivalent quantile from a t -distribution with $n - 2$ degrees of freedom (if n is small or moderate).

Confidence interval for β_1

- Confidence intervals: $\beta_1 = 1.5$ is **known** (left) vs. β_1 is **unknown** (right)



- If β_1 is **known**, 95% of these CIs contain β_1 , and in addition, we know CIs that contain β_1 .
- In practice, β_1 is **unknown**. Therefore, we cannot tell whether a particular CI contains unknown β_1 or not. However, the percentage of CIs that contain unknown β_1 remains the same: 95%.

Hypothesis test for β_1

- **Hypotheses:** $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$

- **Test statistic:**

$$t = \frac{\hat{\beta}_1}{\hat{\sigma} \sqrt{(\mathbf{X}^T \mathbf{X})_{[2,2]}^{-1}}}, \quad \text{where } \hat{\sigma} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **p-value:** $2P(T_{n-2} > |t|)$
 - ① If H_0 is true, the t -test statistic has a t -distribution with $n - 2$ degrees of freedom, denoted by T_{n-2} , for small or moderate n .
 - ② If n is sufficiently large, then the t -test statistic has a $N(0, 1)$ distribution.
 - ③ The p -value can be approximated by $2P(Z > |t|)$ if n is sufficiently large.
- **Decision rule:** Reject H_0 if $p\text{-value} < \alpha$.

Model comparison | Motivation

- Suppose $(x_{i1}, \dots, x_{ip}, y_i)$'s are a random sample of size n with p predictors $X_1, \dots, X_p$ and response Y .
- Multiple linear regression models assume a linear relation between $X_1, \dots, X_p$ and Y :

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i, \quad \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

- Hypothesis test for a single regression coefficient, $H_0 : \beta_j = 0$, can be done using the t -test statistic $t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)} = \frac{\hat{\beta}_j}{\hat{\sigma} \sqrt{(\mathbf{X}^T \mathbf{X})^{-1}_{[j+1, j+1]}}}$.
- **Issue:** How can we test multiple regression coefficients, such as $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$, at the same time?

Nested models

- Consider the following three models:

$$M1 : Y = \beta_0 + \beta_1 X_1 + \epsilon$$

$$M2 : Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

$$M3 : Y = \beta_0 + \beta_1 X_1 + \beta_3 X_3 + \beta_4 X_4 + \epsilon$$

- Nested** models:

- 1 M1 and M2 are **nested**, because M2 with $\beta_2 = 0$ is M1.
- 2 M1 and M3 are also **nested**, because M3 with $\beta_3 = 0$ and $\beta_4 = 0$ is M1.
- 3 However, M2 is **not nested** to M3.

- Comparison of nested models:**

- 1 Is M3 (complex model) **much better**¹ than M1 (simple model)?
- 2 Is M2 (complex model) **much better** than M1 (simple model)?

¹“much better” = “significantly better” = “better by more than chance alone”

Statistical test for nested models | Sketch of theory

- Compare the sum of squared residuals $SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ of simple & complex models.

- If the complex model M3 is better than the simple model M1, then

$$SSR_{\text{complex}} < SSR_{\text{simple}} \Rightarrow SSR_{\text{simple}} - SSR_{\text{complex}} > 0 \Rightarrow SSR_{\text{diff}} > 0$$

where $SSR_{\text{diff}} = SSR_{\text{simple}} - SSR_{\text{complex}}$.

- We compare the mean squared residuals (MSR): $MSR_{\text{complex}} = SSR_{\text{complex}} / (n - 4)$ and $MSR_{\text{diff}} = SSR_{\text{diff}} / 2$. Here, $\frac{SSR_{\text{complex}}}{\sigma^2} \sim \chi^2_{n-4}$ and $\frac{SSR_{\text{diff}}}{\sigma^2} \sim \chi^2_2$. Therefore, $E\left(\frac{SSR_{\text{complex}}}{n-4}\right) = \sigma^2$ and $E\left(\frac{SSR_{\text{diff}}}{2}\right) = \sigma^2$. Note: SSR_{complex} has df = $n - 4$,² SSR_{simple} has df = $n - 2$,³ and SSR_{diff} has df = 2.⁴

- Therefore, if M3 is better than M1, then we have

$$SSR_{\text{diff}} > 0 \Rightarrow F = \frac{MSR_{\text{diff}}}{MSR_{\text{complex}}} > 1$$

- F value of much greater than 1 (or equivalently, smaller p-value) is very strong evidence against the simple model M1.

²because M3 estimates four regression coefficients $\beta_0, \beta_1, \beta_3, \beta_4$

³because M1 estimates two regression coefficients β_0, β_1

⁴ $SSR_{\text{diff}} = SSR_{\text{simple}} - SSR_{\text{complex}}$ has df = 2, because $(n - 2) - (n - 4) = 2$, the difference in the number of parameters for two models in comparison

F-test for nested models | Example - M1 vs. M3

- **Hypotheses:** H_0 : M1 (i.e, M3 with $\beta_3 = 0$ and $\beta_4 = 0$) vs. H_a : M3
- **Test statistic:**

$$F = \frac{MSR_{\text{diff}}}{MSR_{\text{complex}}} = \frac{SSR_{\text{diff}}/2}{SSR_{\text{complex}}/(n-4)}$$

- **p-value:** $P(F_{2,n-4} > F)$
 - 1 If H_0 is true, the F -test statistic has an F -distribution with 2 and $n - 4$ degrees of freedom (df), denoted by $F_{2,n-4}$.
 - 2 Note SSR_{complex} has df = $n - 4$, because M3 estimates four regression coefficients $\beta_0, \beta_1, \beta_3, \beta_4$.
 - 3 Similarly, SSR_{simple} has df = $n - 2$, because M1 estimates two regression coefficients β_0, β_1 .
 - 4 Therefore, $SSR_{\text{diff}} = SSR_{\text{simple}} - SSR_{\text{complex}}$ has df = 2, because $(n - 2) - (n - 4) = 2$, the difference in the number of regression coefficients for two models in comparison.
- **Decision rule:** Reject H_0 if $p\text{-value} < \alpha$.

F-test for nested models | General case

- Suppose a simple model contains p_{simple}^* regression coefficients and a complex model has p_{complex}^* regression coefficients. Let $p_{\text{diff}}^* = p_{\text{complex}}^* - p_{\text{simple}}^*$.

- Hypotheses:** H_0 : Simple model vs. H_a : Complex model

- Test statistic:**

$$F = \frac{\text{MSR}_{\text{diff}}}{\text{MSR}_{\text{complex}}} = \frac{\text{SSR}_{\text{diff}} / p_{\text{diff}}^*}{\text{SSR}_{\text{complex}} / (n - p_{\text{complex}}^*)}$$

- p-value:** $P(F_{p_{\text{diff}}^*, n - p_{\text{complex}}^*} > F)$

- If H_0 is true, the F -test statistic has an F -distribution with p_{diff}^* and $n - p_{\text{complex}}^*$ degrees of freedom (df), denoted by $F_{p_{\text{diff}}^*, n - p_{\text{complex}}^*}$.
- $\text{SSR}_{\text{complex}}$ has df = $n - p_{\text{complex}}^*$, because Complex model estimates p_{complex}^* regression coefficients.
- $\text{SSR}_{\text{simple}}$ has df = $n - p_{\text{simple}}^*$, because Simple model estimates p_{simple}^* regression coefficients.
- Therefore, $\text{SSR}_{\text{diff}} = \text{SSR}_{\text{simple}} - \text{SSR}_{\text{complex}}$ has df = p_{diff}^* , because $(n - p_{\text{simple}}^*) - (n - p_{\text{complex}}^*) = p_{\text{complex}}^* - p_{\text{simple}}^* = p_{\text{diff}}^*$, the difference in the number of regression coefficients for two models in comparison.

- Decision rule:** Reject H_0 if $p\text{-value} < \alpha$.

F-test for nested models | General case

- We often summarize the previous results in the following ANOVA table.

Source	SSR	df	MSR	F	p-value
Difference	SSR_{diff}	p_{diff}^*	$MSR_{\text{diff}} = SSR_{\text{diff}} / p_{\text{diff}}^*$	$\frac{MSR_{\text{diff}}}{MSR_{\text{complex}}}$	$P(F_{p_{\text{diff}}^*, n-p_{\text{complex}}^*} > F)$
Complex	SSR_{complex}	$n - p_{\text{complex}}^*$	$MSR_{\text{complex}} = SSR_{\text{complex}} / (n - p_{\text{complex}}^*)$		
Simple	SSR_{simple}	$n - p_{\text{simple}}^*$			

Example | Number of species

- **Example 3.1:** Refer to Section 3.4 of the online textbook.⁵

- 1 First, we consider the full (complex) model that predicts the number of species as a function of elevation, area, nearest, scruz, and adjacent. This model has a total of $p_{\text{complex}}^* = 6$ regression coefficients $\beta_0, \beta_1, \dots, \beta_5$ (one for each predictor plus the y -intercept):

$$\text{Species}_i = \beta_0 + \beta_1 \text{Area}_i + \beta_2 \text{Elevation}_i + \beta_3 \text{Nearest}_i + \beta_4 \text{Scruz}_i + \beta_5 \text{Adjacent}_i + \epsilon_i, \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

- 2 The first test we might want to do is to test if any of the predictors is significant. That is,

$$H_0 : \text{all predictors are not significant } (\beta_1 = \dots = \beta_5 = 0)$$

$$H_a : \text{at least one of the predictors is significant}$$

- 3 If H_0 is true, the complex model can be expressed as a reduced (simple) model:

$$y_i = \beta_0 + \epsilon_i$$

- 4 F-tests are a rigorous way to compare nested models and decide if a complex model significantly explains more variability in responses than a simple model to support additional predictors in the complex model.

⁵Data: gala in the faraway package.

Example | F -test for nested models

- R scripts to perform a model comparison F -test:

```
1 # Fit the full (complex) model
2 m_full = lm( Species ~ Area + Elevation + Nearest + Scrub + Adjacent, data=gala )
3
4 # Fit a reduced model (no predictors, intercept only)
5 m_red1 = lm( Species ~ 1, data=gala )
6
7 # F-test by using the anova() command
8 anova( m_red1, m_full )
```

- F -test results from the anova() command:

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	29	381081.37	NA	NA	NA	NA
2	24	89231.37	5	291850	15.69941	6.837893e-07

Example | ANOVA table and conclusion

- ANOVA table:

Source	SSR	df	MSR	F	<i>p</i> -value
Difference	291850.00	5	58370.000	15.699	0.00000068
Complex	89231.37	24	3717.974		
Simple	381081.37	29			

- MSRs, *F*-test statistic, and *p*-value are calculated as follows:

$$MSR_{\text{diff}} = \frac{SSR_{\text{diff}}}{p_{\text{diff}}^*} = \frac{SSR_{\text{simple}} - SSR_{\text{complex}}}{(n - p_{\text{simple}}^*) - (n - p_{\text{complex}}^*)} = \frac{381081.37 - 89231.37}{29 - 24} = 58370,$$

$$MSR_{\text{complex}} = \frac{SSR_{\text{complex}}}{n - p_{\text{complex}}^*} = \frac{89231.37}{24} = 3717.974,$$

$$F = \frac{MSR_{\text{diff}}}{MSR_{\text{complex}}} = \frac{58370}{3717.974} = 15.699,$$

$$p\text{-value} = P(F_{5,24} > 15.699) = 0.00000068$$

- Conclusion:** Because *p*-value = 0.00000068 is smaller than $\alpha = 0.05$, we reject H_0 . This implies that at least one predictor is significant at $\alpha = 0.05$ level of significance.